Assignment #1 CLO-1 BT level C2

Date of submission: 15/10/2026

Question # 1 Show that

$$\vec{d} \times (\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c}) = [\vec{a} \quad \vec{b} \quad \vec{c}](\vec{a} \cdot \vec{d})$$

Question # 2

Suppose that $\vec{u},\ \vec{v}\ and\ \vec{w}$ are vectors in 3-space such that $\left|\vec{u}\right| = 5,\ \vec{u} \cdot \vec{v} = 7\ and\ \vec{u} \cdot \vec{w} = -3$ then compute

(a) $\vec{u} \cdot \vec{u}$ (b) $\vec{v} \cdot \vec{u}$ (c) $\vec{u} \cdot (\vec{v} - \vec{w})$ (d) $\vec{u} \cdot (2\vec{w})$

Question # 3

a) Use the fact that $\vec{u} \cdot \vec{v} = \left|\vec{u}\right| \left|\vec{v}\right| cos\,\theta$ to show that the inequality $\left|\vec{u} \cdot \vec{v}\right| \leq \left|\vec{u}\right| \left|\vec{v}\right|$ holds for any vectors $\vec{u}$ and $\vec{v}$.

b) Under what circumstances, if any, does $\left|\vec{u} \cdot \vec{v}\right|$ equal $\left|\vec{u}\right| \left|\vec{v}\right|$? Give reasons for your answers.

Question # 4

Compute

$$\frac{\vec{b} \times \vec{c}}{[\vec{a} \quad \vec{b} \quad \vec{c}]} \cdot \frac{\vec{c} \times \vec{a}}{[\vec{a} \quad \vec{b} \quad \vec{c}]} \times \frac{a \times b}{[\vec{a} \quad \vec{b} \quad \vec{c}]}$$

Question # 5

Compute an equation for the plane through $A(1,\ 1,-1),\ B(2,0,2)$ and $C(0,-2,1)$.